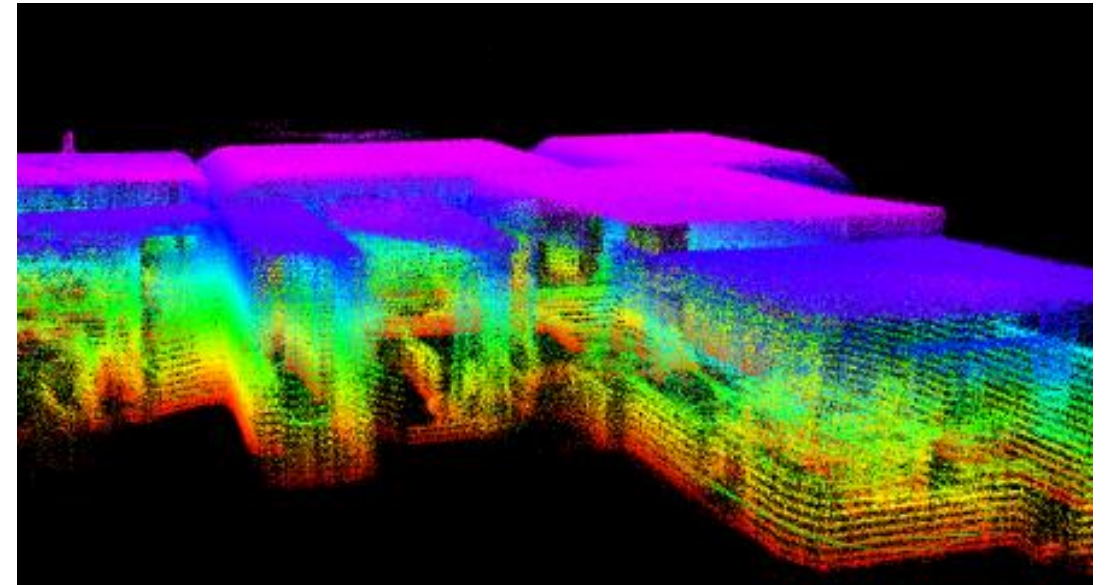
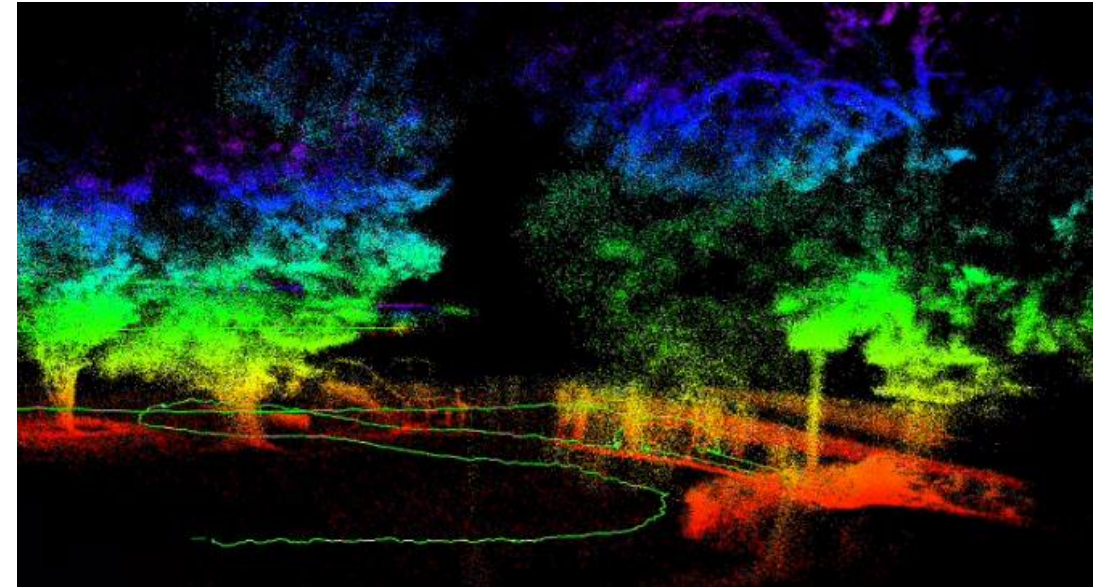


INTERNAL PRESENTATION (GROUP 4) LIDAR SCOUT ROVER

- Zabdiel Montano
- Randy Rodriguez
- Nicholas Boggs
- Andrew Patino



Introduction

Background / Motivation

- Some environments are unsafe or difficult for direct human inspection.
- Many robotic systems are expensive or limited to specific tasks.
- A low-cost ground robot can help perform remote inspection.

Purpose

- The LiDAR Scout Rover remotely maps and monitors hazardous areas.
- It improves situational awareness before people enter.

Use Cases

- Collapsed structures
- Surveying/Patrolling areas
- Gas leak sites
- Confined spaces

Key Features

- LiDAR-based real-time mapping
- Camera and gas sensor monitoring
- Ground-based multi-terrain mobility
- Modular ROS 2 architecture
- Affordable open-source platform

Project Management: Schedule

- Semester 1 – Core Rover Development

- **Wk1-2:** Researched optimal mobility design and compatible hardware/software.
- **Wk3-4:** Created proposal and finalized budget.
- **Wk5-6:** Ordered major components (LiDAR, sensors, chassis) and configured Raspberry Pi 5.
- **Wk7-8:** Installed Ubuntu + Ros 2 dependencies; assembled chassis and wiring schematic.
- **Wk9-10:** Mounted hardware, verified LiDAR operation and stable power.
- **Wk11-12:** Integrated ROS 2 communication and began motor testing.
- **Wk13-14:** Finalized wiring; achieved ROS 2 motor control.
- **Wk15:** Delivered end-of-semester report and presentation.

- Semester 2 – Autonomy & Field Testing

- **Wk16-17:** Refined ROS 2 motor control and navigation response.
- **Wk18-19:** Linked LiDAR mapping with motor control for path following.
- **Wk20-21:** Enhanced web dashboard and Wi-Fi connectivity
- **Wk22-23:** Added camera & gas sensors; validated data accuracy
- **Wk24-25:** Re-designed and printed improved chassis/mounts.
- **Wk26-27:** Performed full-system testing and data collection
- **Wk28-29:** Compiled final report and demo materials.
- **Wk30:** Presented final project and submitted documentation.

Design/Implementation

Design

- Tracked rover platform with LiDAR, IMU, camera, gas sensor, and onboard compute
- ROS 2 node-based architecture for modular integration
- Point-LIO odometry feeding Nav2 navigation
- 3D printed front plate/skid plate using Fusion360

Implementation

- PCL (Point Cloud Library) used for obstacle extraction from the 3D point cloud
- Custom advanced_proximity_guard node used for close-range avoidance
- nav2_costmap_2d used for 2D navigation costmaps
- Arduino programmed for low-level rover control
- Website interface built for monitoring and manual control

Project Management: Budget

Component	Model/Example	QTY	Unit Cost (\$)	Subtotal (\$)
Tracked Chassis	Premium Large Metal 4WD Shock Absorption Robot Tank Car Chassis Kit (includes frame, tracks, motors)	1	100.00	100.00
Raspberry Pi 5 Kit (8 Gb)	RasTech Pi 5 kit (case, cooler, PSU, SD)	1	64.99	64.99
Motor Driver	L293D Motor Drive Shield Expansion Board	1	7.29	7.29
ELEGOO Mega Board	Elegoo MEGA 2560 R3 Board	1	44.99	44.99
LiDAR	Unitree 4D LiDAR L1 (L1PM 20m)	1	249.00	249.00
Camera	Arducam IMX219 8MP Autofocus USB Camera Module	1	29.99	29.99
Gas Sensors + ADC	MQ kit (MQ-2/7/135...) + MCP3008	1	24.99	24.99
Batteries + Power	6S LiPo (22.2 V) + XT90 anti-spark + 5 V/5 A buck + 12V LiPo Battery	1	103.00	103.00
Networking	BrosTrend 650Mbps USB WiFi Adapter (Linux & Pi compatible, long range antenna)	1	23.99	23.99
Mounts & Misc	3D-printed brackets, wiring, fuses, connectors	1	40.00	40.00
Contingency & Incidentals	Extra parts, bolts, tools used	1	80.00	80.00
Total				768.30

Student Roles & Responsibilities

Nicholas Boggs

Role: Mechanical Integration and Power Distribution

1st Semester

- Assembled tank chassis and integrated the tracked rover platform
- Managed power distribution wiring between the battery, motors, converters, and onboard electronics
- Helped organize wiring layout for cleaner integration and easier troubleshooting

2nd Semester

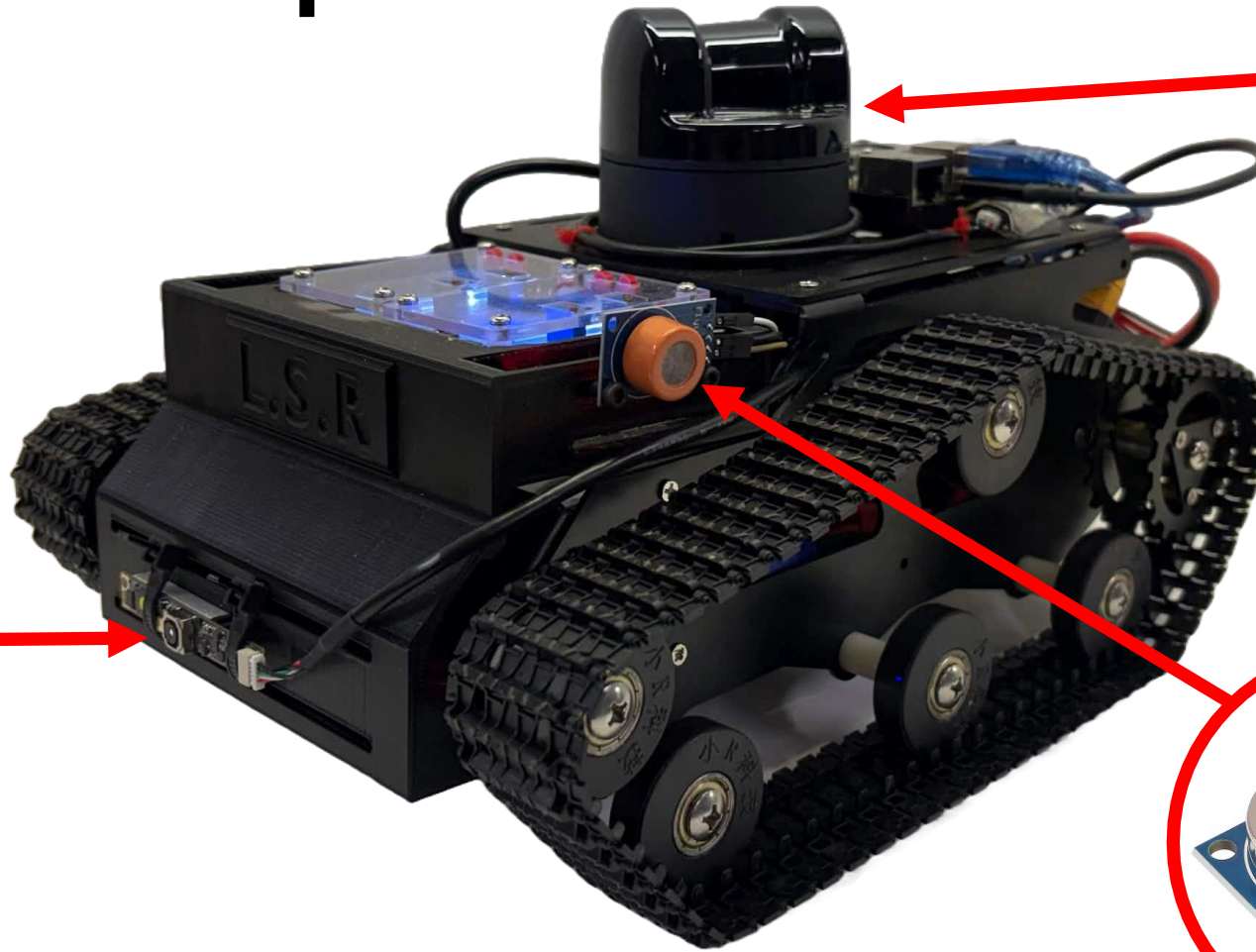
- Created Fusion 360 models for mounting and integration of rover components
- Continued supporting chassis integration and overall system assembly
- Assisted with refining power routing and hardware placement for improved reliability

Rover Specifications



-
- Size: 32x23x20 (cm)
 - Weight: 3.05 Kg
 - Connection Range: $\approx$ 40m on 5GHz Ethernet Connection
 - Rover Battery Life: 5-6 Hours
 - LiDAR Battery Life: 2-3 Hours
 - 3D-printing filament used: PLA

Exterior Components



Unitree 4D
LiDAR L1
(L1PM 20 m)

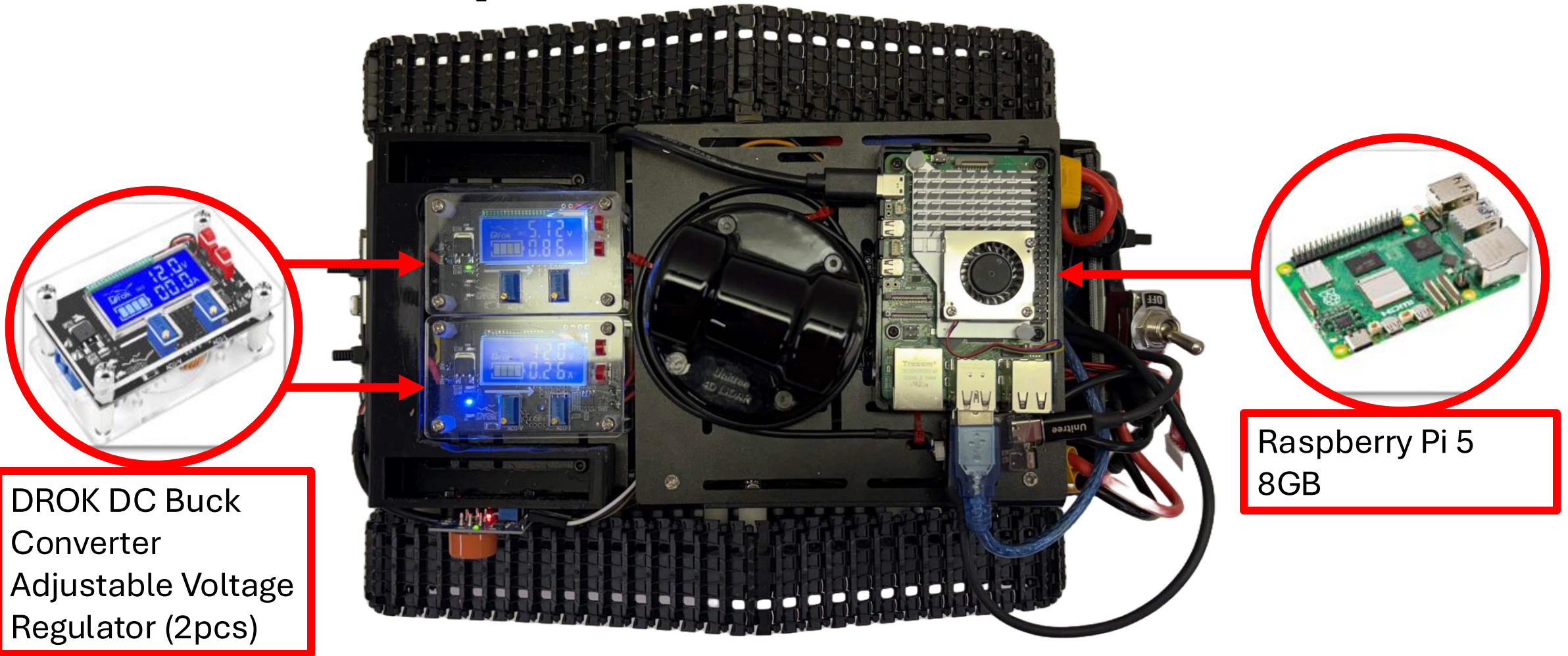


Arducam
IMX219 8MP
Autofocus USB
Camera Module



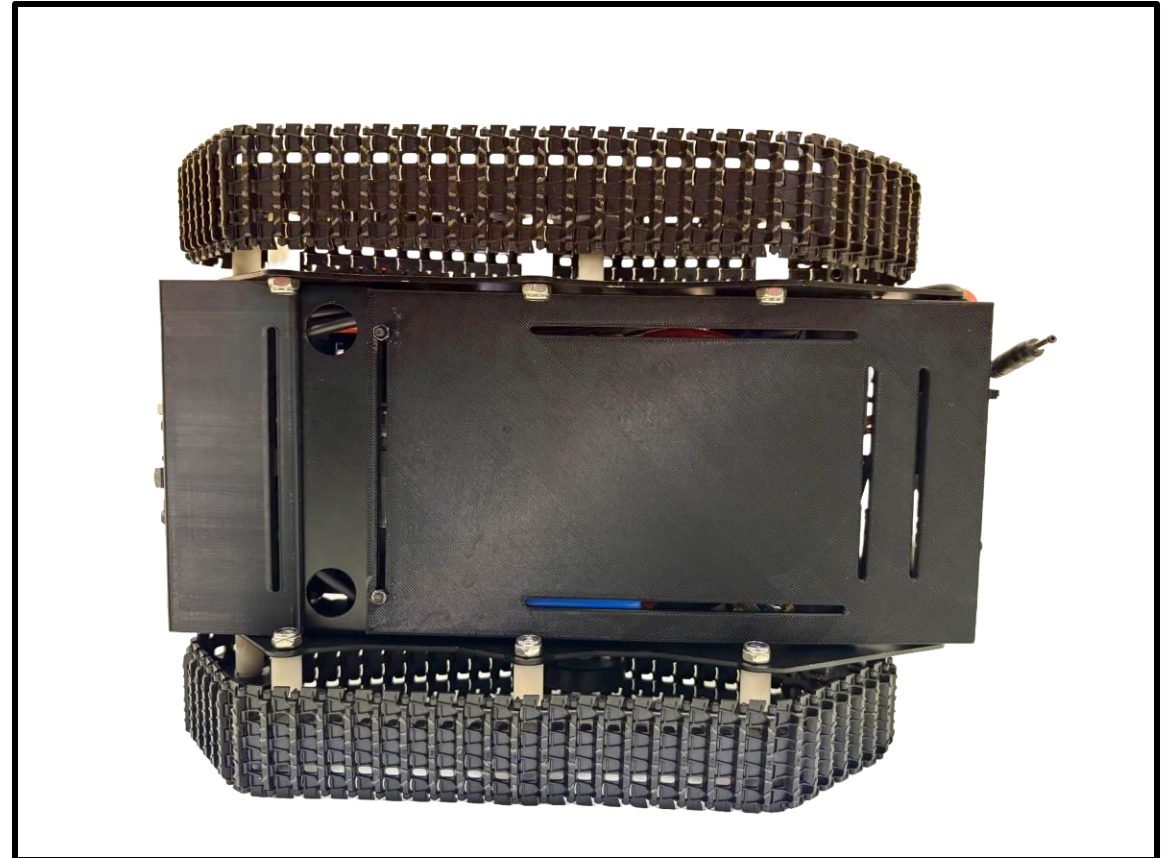
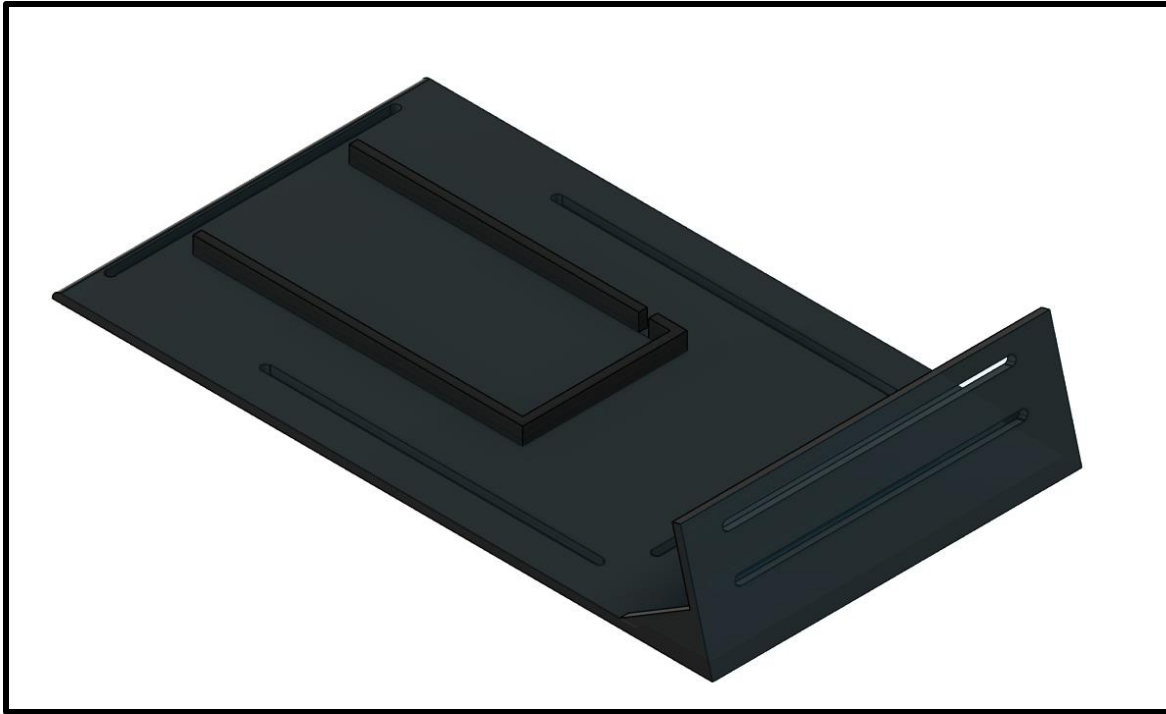
ACEIRMC (MQ-3)/Lot Gas
Detection
Sensor Module
(Alcohol Gas
Sensor)

Exterior Components



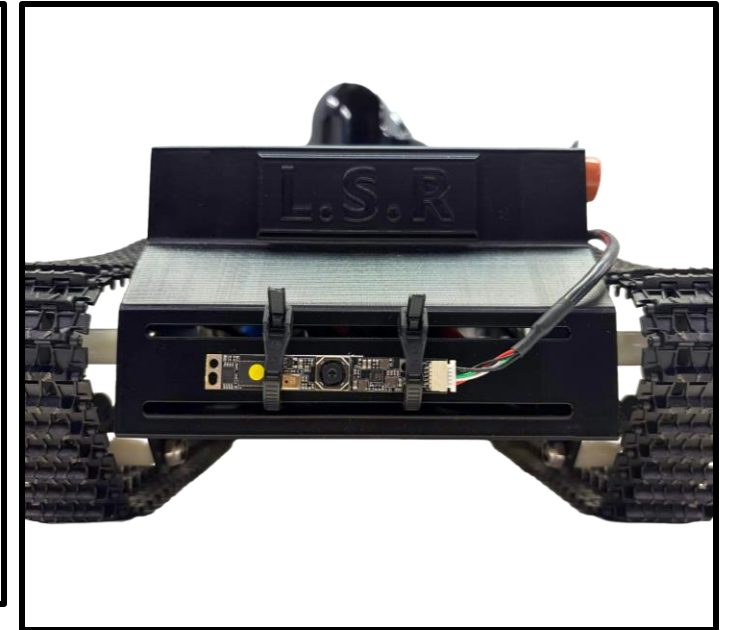
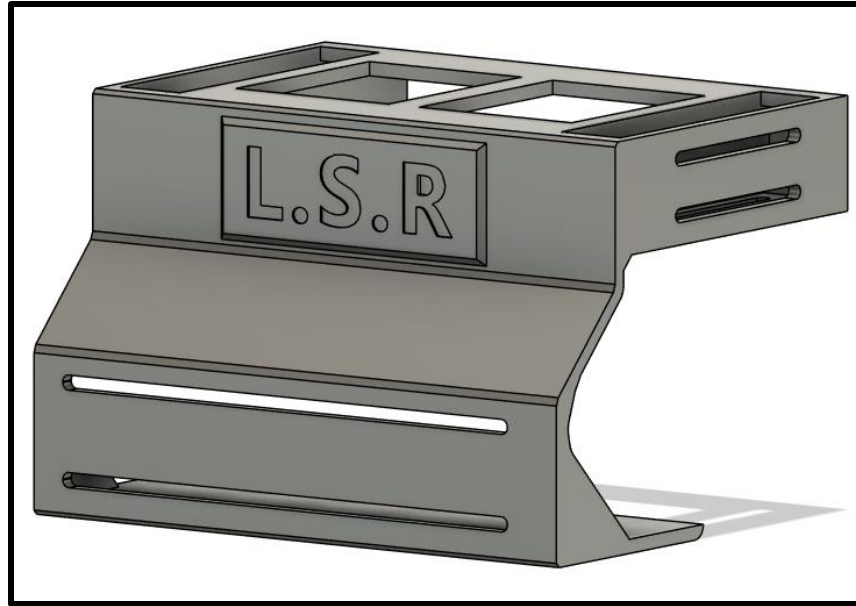
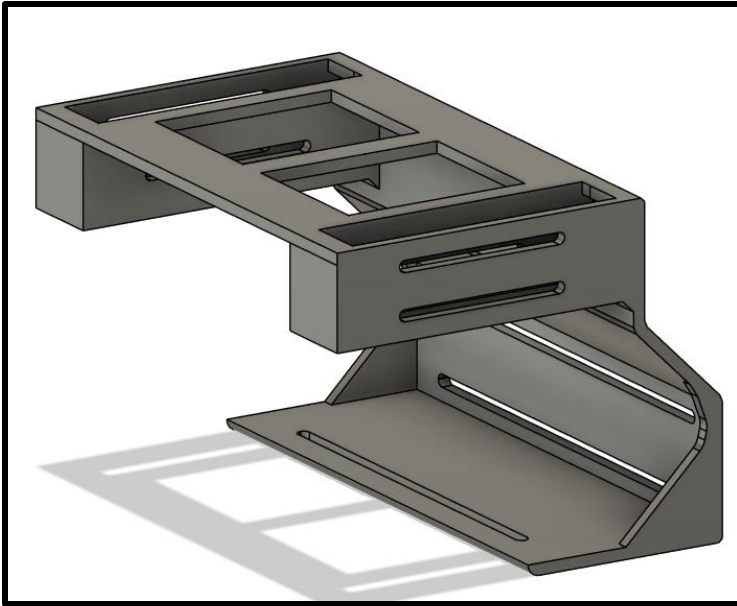
3D Printed Parts

Skid Plate



3D Printed Parts

Front Plate



Student Roles & Responsibilities

Andrew Patino

Role: Electrical Systems Designer and Integration

1st Semester

- Installed Ubuntu OS onto the Raspberry Pi 5.
- Documented wiring connections to allow for debugging.
- Assisted with power distribution between all components.

2nd Semester

- Implemented hardware connections between microcontroller, sensors, and power systems into a cohesive embedded platform.
- Verified circuit functionality through testing and debugging, resolving wiring and connectivity issues.
- Organized and documented schematics and wiring layouts to support system replication and troubleshooting.

Interior Components



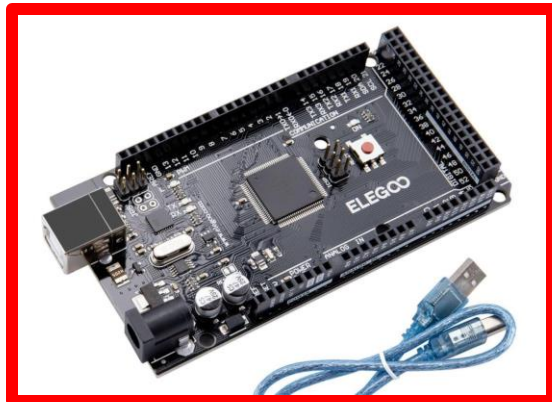
22.2V 6Ah
LiPo Battery



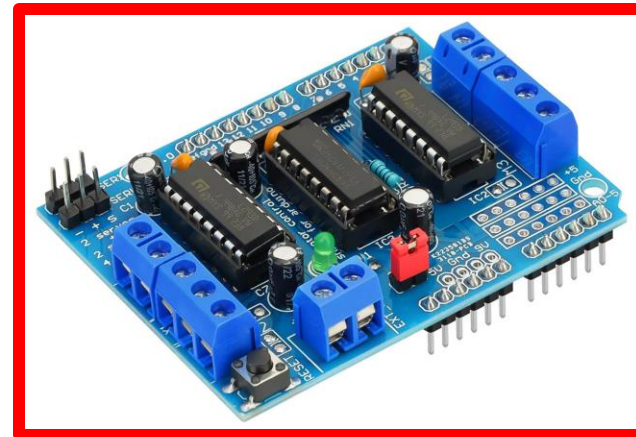
12V 2.6 Ah
LiPo
Battery



ACEIRMC(MQ-2)/Lot Gas
Detection Sensor
Module
(Gas & Smoke
Sensor)



ELEGOO
MEGA R3
Board ATmega
2560



L293D DC Motor
Drive - Stepper
Motor Control
Shield Module
Expansion Board

Camera/Gas Sensors

```
--- SCOUT ROVER: GAS SENSOR SYSTEM ---
MQ-2 (A11): Flammable Gas & Smoke Sensor no specified range |
Warming up... Please wait 60 seconds for sensor heaters.
Warmup: 60s remaining...
Warmup: 50s remaining...
Warmup: 40s remaining...
Warmup: 30s remaining...
Warmup: 20s remaining...
Warmup: 10s remaining...
System Ready. Monitoring Started.
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
MQ-2: 71 [0.35V] | MQ-3: 275 [1.34V]
```

MQ-2: Combustible Gas & Smoke Sensor

- The MQ-2 is a versatile Metal Oxide Semiconductor (MOS) sensor. It contains a sensing element made of Tin Dioxide (SnO_2), which has low electrical conductivity in clean air.
- Target Gases: Smoke, LPG (Liquefied Petroleum Gas), Butane, Propane, Methane, Alcohol, and Hydrogen.
- How it works: When combustible gases are present, the sensor's conductivity increases. The module converts this change in resistance into an analog voltage that the Arduino Mega reads via the Analog-to-Digital Converter (ADC) on pin A11. Best Use Case: Fire detection and general leak monitoring.

MQ-3: Alcohol Gas Sensor

- Like the MQ-2, the MQ-3 is specifically chemically "tuned" to be sensitive to alcohol vapors while being less reactive to smoke or LPG.
- Target Gases: Alcohol, Ethanol, and various solvent vapors.
- How it works: It uses a heater-driven chemical reaction to detect ethanol concentration in the air. This is connected to pin A9. This is connected to pin A9.
- Best Use Case: Detecting chemical spills or breathalyzer-style environmental monitoring.



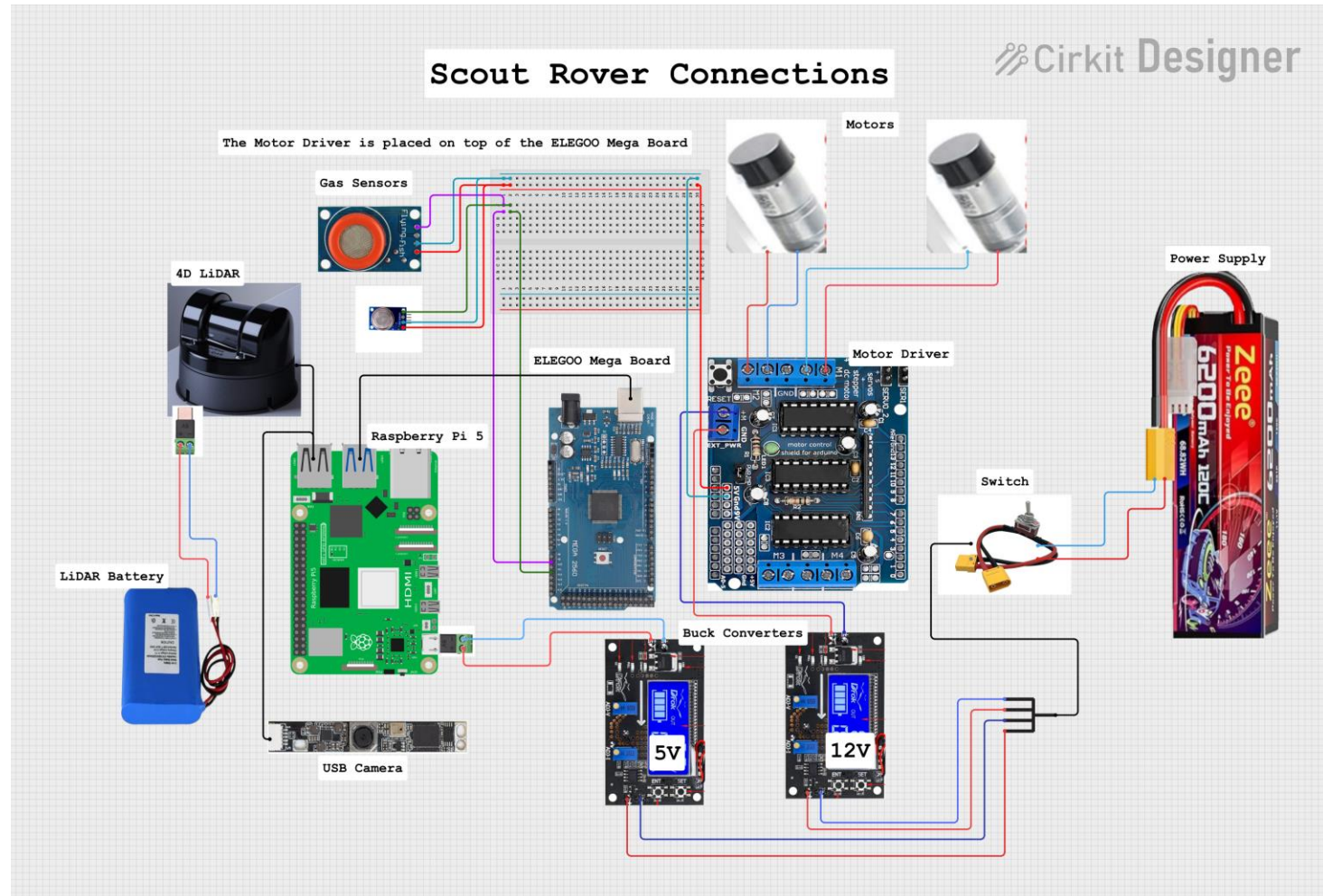
RVIZ 2 /image from webcam

Command:

```
ros2 run v4l2_camera v4l2_camera_node --ros-args -p
video_device:="/dev/video0" -p image_size:="[640,480]"
```

Electrical System Overview

- Power supply splits between our buck converters 5V and 12V.
- **5V** gets sent to the **Raspberry Pi 5** which delegates voltage to the **USB Camera**.
- **12V** goes to the **Motor Drivers** which sends the voltage to the **Motors** and the **Gas Sensors**.
- **12V** gets sent to the **LiDAR** with its own battery.



Student Roles & Responsibilities

Zabdiel Montano

Role: Web Interface and Dashboard Developer

- Developed the rover's website dashboard for monitoring and control.
- Integrated live camera, telemetry, teleoperation, and ROS topic views.

Website

- The Overview page is the main dashboard for live rover monitoring.
- It combines camera feed, connection status, control source, goal distance, gas sensor data, and obstacle activity in one screen.
- It also includes the on-page teleoperation controls and the local navigation view centered on base_link.
- This page is used for quick situational awareness during rover operation.

The screenshot displays the 'LIDAR SCOUT ROVER' Overview page. The interface is dark-themed and organized into several sections:

- Left Sidebar:** Contains navigation links for Overview, Stack, Navigation, Controller, and Topics. The Overview link is active.
- Top Bar:** Shows status indicators for ROS BRIDGE OFFLINE, CAMERA WAITING, and MODE IDLE.
- Overview Section:** Provides a live camera, rover pose, command stream, and on-page teleop. It includes four sub-widgets:
 - ROS BRIDGE:** Status 'Offline', 'Waiting for rosbrowser websocket'.
 - CONTROL SOURCE:** Status 'Idle', 'No recent velocity publisher'.
 - GOAL DISTANCE:** Status '—', 'No active Nav2 path'.
 - OBSTACLE MARKERS:** Status '0', 'No cluster markers right now'.
- GAS SENSOR:** Status '—', 'waiting on /gas_raw'.
- CAMERA SCREEN:** Base topic: /image_raw with browser video from /image_raw/compressed. It shows a 'Camera feed' section with 'NO STREAM' and a 'CAMERA WAITING' message: 'Waiting for /image_raw/compressed through rosbrowser, or optionally web_video_server.'.
- CONTROLLER:** Publishes manual teleop to /cmd_vel, teleop when rosbrowser is connected. It features a directional pad with 'STOP' in the center, a 'THROTTLE' slider set to 0.45 m/s, and an 'ANGULAR SPEED' set to 0.90 rad/s. Keyboard shortcuts are noted as W, A, S, D, and Space.
- LOCAL NAVIGATION VIEW:** Robot-centered view from base_link using the local plan, obstacle markers, and active safety zones. It shows a grid with a robot icon and a 'forward' arrow. Status at the bottom: 'base_link centered | yaw: 0.0° | local plan: 0 poses'.
- LIVE COMMANDS:** Current output on /cmd_vel and planner state. It lists Linear X (0.00 m/s), Angular Z (0.00 rad/s), Pose X (0.00 m), Pose Y (0.00 m), Yaw (0.0°), Global plan length (0 poses), and Gas raw (—).

Website

- The Stack page summarizes the rover’s active runtime components and system layout.
- It shows the major ROS 2 topics and subsystems used for sensing, odometry, navigation, safety, and final motor commands.
- It also includes rover geometry and connection notes for the current Point-LIO and Nav2 bringup.
- This page helps explain how the rover software pipeline is organized.

LIDAR SCOUT ROVER

LSR · ROS 2 Jazzy · Nav2 + Point-LIO

Overview

Stack

Navigation

Controller

Topics

STACK

Current LSR runtime components and connection assumptions.

Rover Stack

RUNTIME COMPONENTS

Updated for the current LSR bringup.

Unitree LIDAR

/unitlidar/cloud

Restamp bridge

/unitlidar/cloud_restamped

Point-LIO odometry

/odom_corrected -> /odom

Proximity guard

/filtered_obstacles · /obstacle_markers

Nav2 planner/controller

/plan · /local_plan

Velocity mux

/cmd_vel_nav · /cmd_vel_avoid · /cmd_vel_teleop

Final motor command

/cmd_vel

VEHICLE GEOMETRY

Current dashboard copy aligned to your latest rover dimensions.

Length

Width

Height

LiDAR mount

LiDAR TF Z

Danger zone

CONNECTION NOTES

This page is connection-ready, but it expects a couple of services to be running on the rover host.

rosbridge_server is needed for topic telemetry and the on-page controller.

The rover bringup should already be bridging Point-LIO odometry into /odom, typically from /odom_corrected, before Nav2 and this dashboard come online.

The camera panel prefers /image_raw/compressed through rosbridge, while web_video_server remains an optional fallback on the same host.

Website

- The Navigation page focuses on rover pose, path state, and active plan information.
- It displays pose values, global and local plan status, goal distance, and current control source.
- This page is used to monitor how Nav2 is planning and updating motion during autonomous operation.
- It gives a quick view of the rover's navigation state without opening RViz.

LIDAR SCOUT ROVER

LSR · ROS 2 Jazzy · Nav2 + Point-CloudIO

Overview

Stack

Navigation

Controller

Topics

NAVIGATION

Robot-centered Nav2 path state, obstacle geometry, and active safety zones.

Navigation

POSE X

0.00 m

from /odom

POSE Y

0.00 m

from /odom

YAW

0.0°

derived from quaternion

PLAN STATE

Live path summaries instead of example goal inputs.

Global plan

waiting

Local plan

waiting

Goal distance

—

Control source

idle

Teleop throttle

0.45 m/s

Teleop turn

0.90 rad/s

CURRENT LIMITS

Frontend display for the rover's present tuning.

Cruise target

Avoid reverse

Avoid turn

Teleop linear

Teleop angular

ROS BR

Website

- The Controller page is focused on manual command control and drive mode awareness.
- It shows whether the rover is currently being driven by navigation, avoidance, manual input, or is idle.
- It also explains how velocity commands flow through `/cmd_vel_nav`, `/cmd_vel_avoid`, `/cmd_vel_teleop`, and final `/cmd_vel`.
- This page is useful for understanding which command source has control of the rover at any moment.

LIDAR SCOUT ROVER
LSR · ROS 2 Jazzy · Nav2 + Point-LIO

Overview

Stack

Navigation

Controller

Topics

CONTROLLER
Manual teleop plus current command-source mode detection.

Control Modes

MODE STATE
Highlighted from the most recent command source seen on the rover.

NAVIGATION
Nav2 is actively driving through `/cmd_vel_nav`.

AVOIDANCE
The proximity guard is overriding with `/cmd_vel_avoid`.

MANUAL
The dashboard controller is publishing to `/cmd_vel_teleop`.

IDLE
No recent velocity source is active.

HOW IT CONNECTS
No more example input blocks. This is wired around your real rover topics.

`/cmd_vel_nav` is treated as autonomous driving.
`/cmd_vel_avoid` is treated as safety override.
`/cmd_vel_teleop` is the controller topic this page publishes.
`/cmd_vel` is the final muxed output the rover motor bridge should consume.

Website

- The Topics page lists the main ROS 2 topics used by the rover system.
- Topics are grouped by perception, navigation, and control for easier monitoring.
- It provides a quick live reference for the current Point-LIO, Nav2, obstacle, and command topics used in the project.
- This page serves as a lightweight runtime topic monitor and integration reference.

LIDAR SCOUT ROVER

LSR · ROS 2 Jazzy · Nav2 + Point-LIO

Overview

Stack

Navigation

Controller

Topics

ROS 2 TOPICS

Real rover topics, updated from your current bringup.

ROS 2 Topics

PERCEPTION

/unilidar/cloud

PointCloud2

—

/unilidar/cloud_restamped

PointCloud2

—

/unilidar/imu

Imu

—

/unilidar/imu_restamped

Imu

—

/filtered_obstacles

PointCloud2

—

/obstacle_markers

MarkerArray

—

/gas_raw

Int32

—

NAVIGATION

/odom

Odometry

—

/plan

Path

—

/local_plan

Path

—

/tf

TFMessage

—

/tf_static

TFMessage

—

/navigate_to_pose/_action/status

GoalStatusArray

—

CONTROL

/cmd_vel_nav

/cmd_vel_avoid

/cmd_vel_teleop

/cmd_vel

/image_raw

/image_raw/compressed

/camera_info

CONNECTION SUMMARY

This page is aligned with the current LSR runtime using Point-LIO, Nav2, and the custom proximity guard.

The topic list above follows the rover bringup you have been testing: restamped LiDAR, Point-LIO odometry bridged into /odom, Nav2 plan topics, proximity guard markers, and muxed velocity commands.

If you later move to RTAB-Map or SLAM Toolbox, we can add a dedicated mapping tab and a real /map section without changing the controller or camera panels.

Student Roles & Responsibilities

Randy Rodriguez

Role: ROS 2 Software and Embedded Systems Developer

- Developed and integrated the ROS 2 (Jazzy) navigation and perception pipeline for the rover.
- Configured and tuned Point-LIO, Nav2, RViz2, TF frames, costmaps, and obstacle handling.
- Implemented and refined custom ROS 2 nodes for short-range obstacle avoidance, command control, camera integration, and gas sensor data handling.
- Programmed the Arduino-based low-level rover control interface for hardware actuation and communication.
- Tested, debugged, and validated system behaviour during mapping, navigation, sensing, and real-time obstacle response.

ROS 2 (Robot Operating System 2)

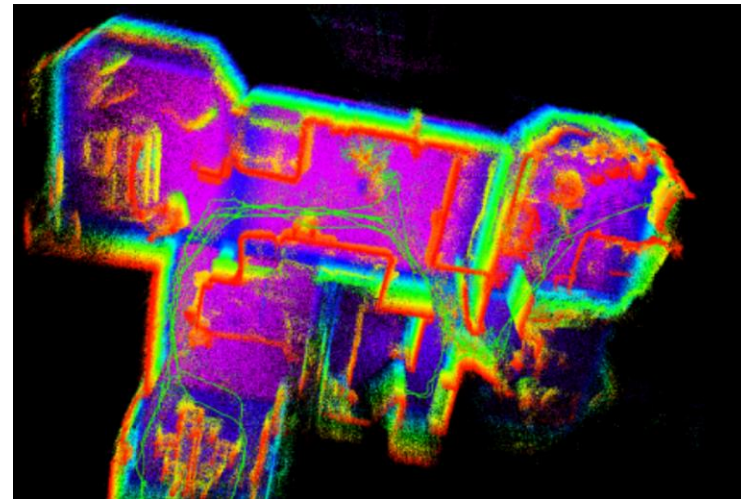
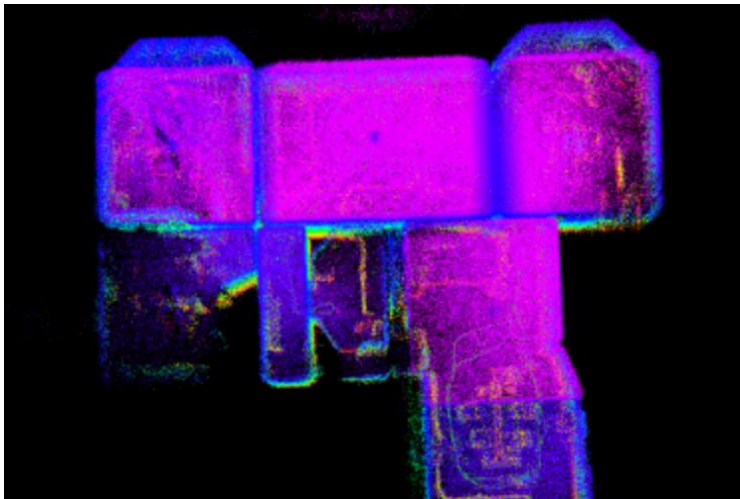
- ROS 2, or Robot Operating System 2, is a software framework used to build robot applications.
- It allows different parts of a robot, such as sensors, navigation, and motor control, to communicate through modular software nodes.
- ROS 2 is widely used in research and industry because it supports real-time communication, scalability, and distributed systems.
- In this project, we used ROS 2 version Jazzy Jalisco



[Source: \[20\]](#)

Point-Lio: Robust High-Bandwidth Lidar-Inertial Odometry

- Point-LIO provides LiDAR-inertial odometry.
- It fuses LiDAR and IMU data to estimate rover pose.
- It also produces the live mapped cloud and path in RViz.
- This is the localization/odometry side of the system.



NAV 2(Navigation 2)

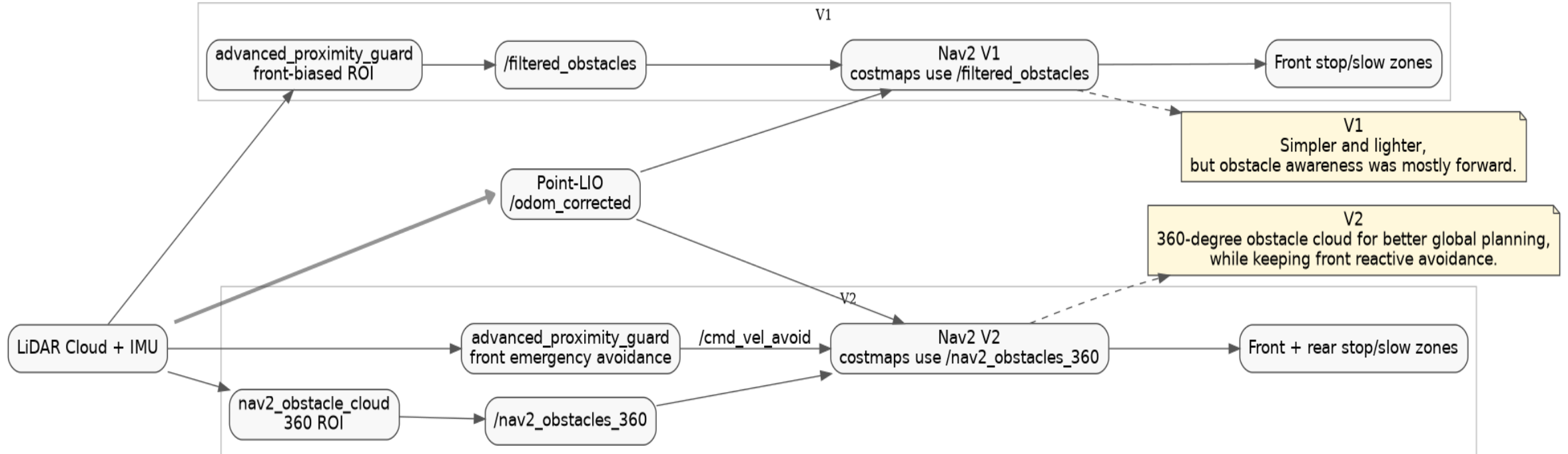
- Nav2 is the ROS 2 navigation framework used to move a robot to a goal while avoiding obstacles.
- In real-world robotics, Nav2 is used for path planning, motion control, and safe navigation in changing environments.



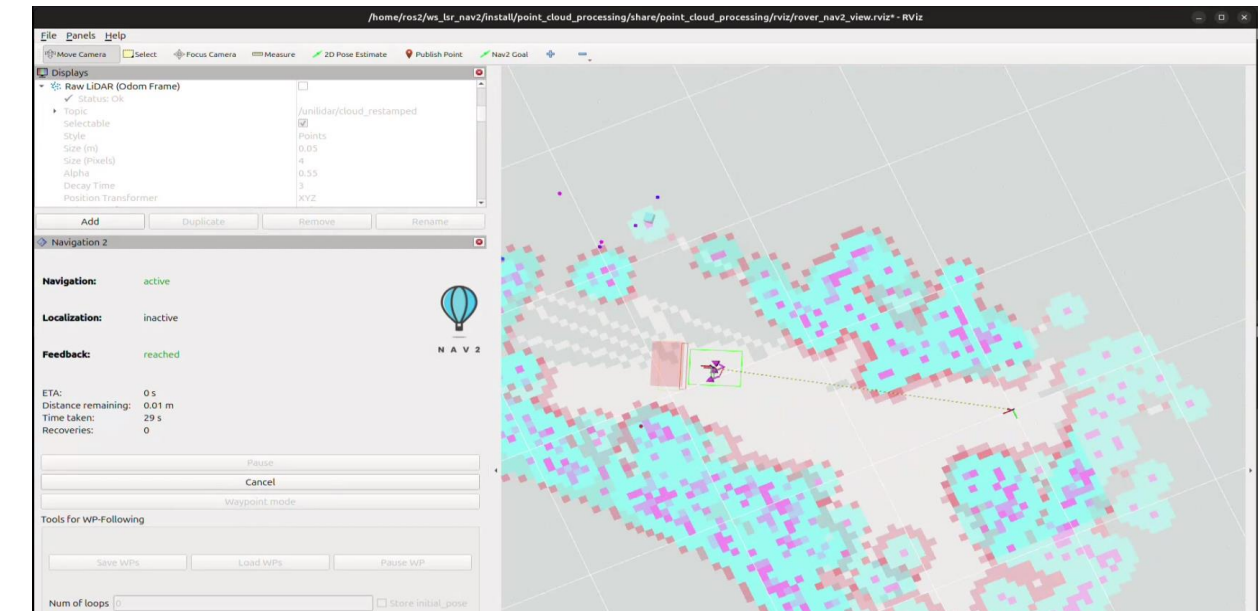
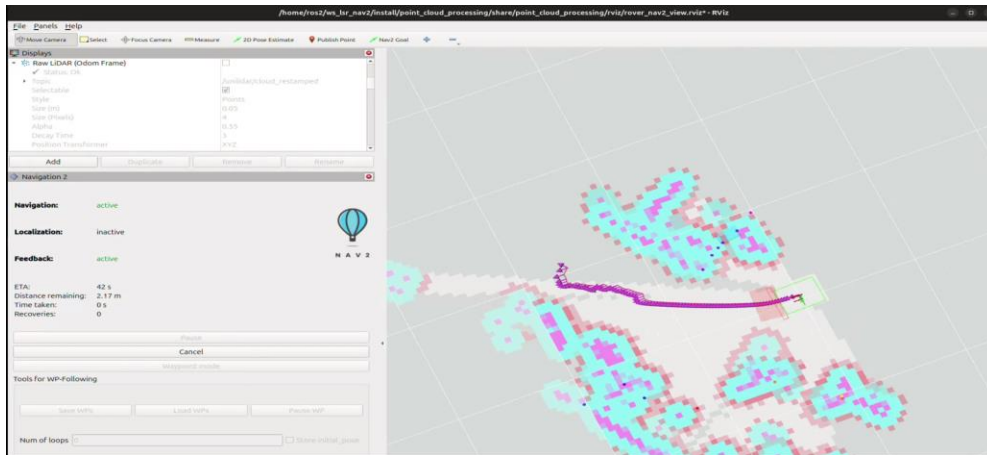
[Source: \[16\]](#)

Point-LIO & NAV 2 System Flowchart

LiDAR Scout Rover - V1 vs V2 Navigation Software



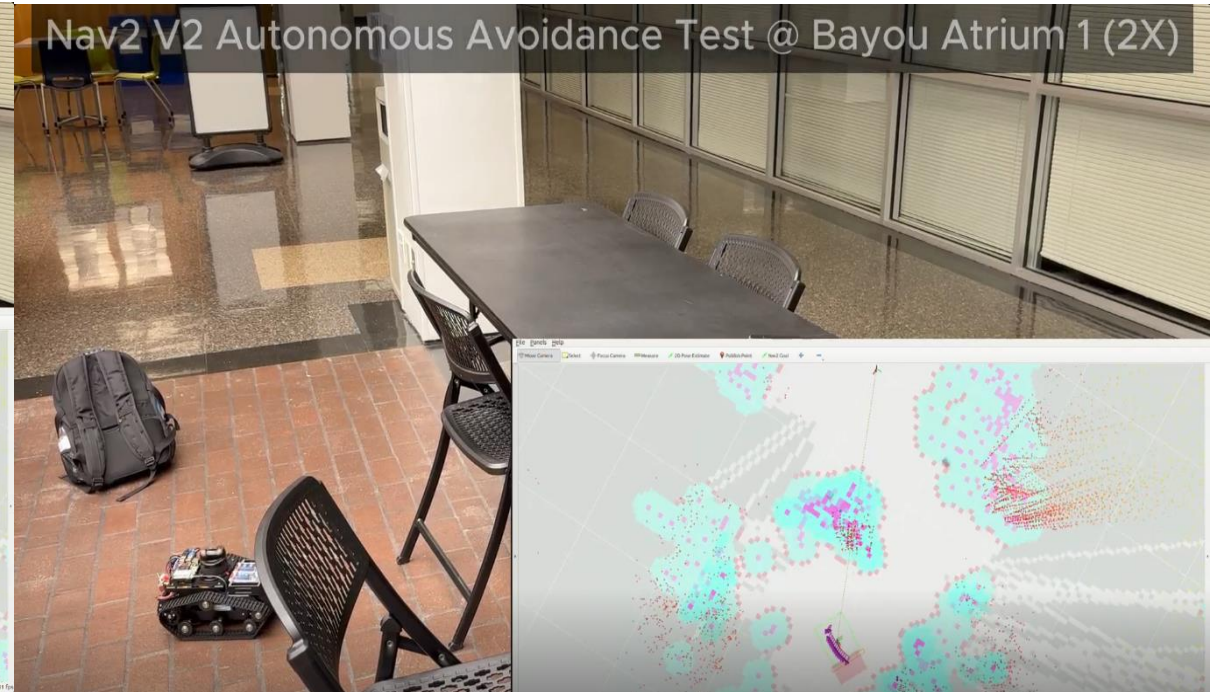
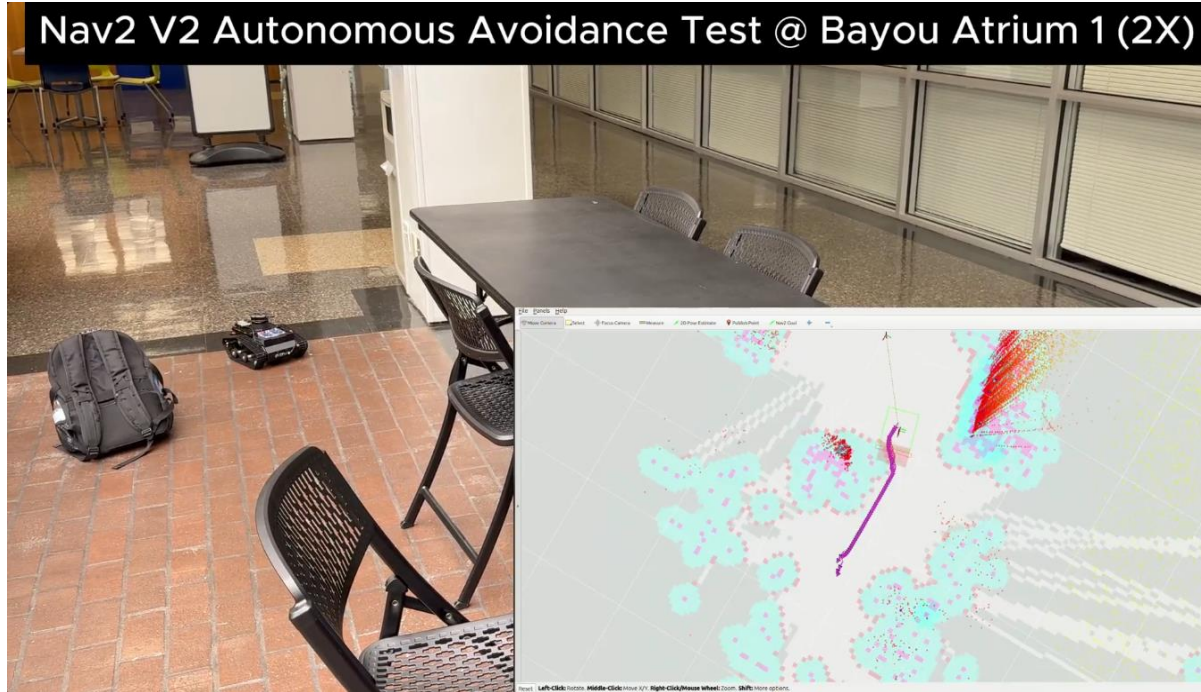
NAV 2(Navigation 2) V1



What RViz2 Shows

- Global Costmap shows the larger navigation area used for path planning.
- Local Costmap shows nearby obstacles around the rover in real time.
- Global Path is the overall route to the goal.
- Local Plan is the short-term path the rover is currently following.
- Footprint, Stop Zone, and Slow Zone show the rover body and safety boundaries.
- Filtered Obstacles shows obstacle points extracted from the LiDAR cloud.

NAV 2(Navigation 2) V2



Relevant Classes and Acquired Skills

Classes

- CENG 4351 – Introduction to Robotics
- CENG 4313 / 4113 – Microprocessor Interfacing & Lab
- CENG 3316 / 3116 – Electronics & Lab
- CENG 3351 / 3151 – Computer Architecture & Lab
- CENG 3264 – Engineering Design & Project Management

Skills

- Digital & Analog circuit design, PCB integration
 - Microcontrollers (programming, interfacing)
 - Embedded systems with Raspberry Pi & ARM-based boards
 - Robotics frameworks (ROS2, RViz2, Point-LIO)
 - Fusion360 modeling & chassis integration
 - Project management & teamwork in engineering projects
-

Conclusion

Each team member contributed equally to the project. We have been gathering all necessary components according to our budget plan and have researched relevant video examples related to our rover. This experience has provided valuable insight into the research, planning, and coordination required for complex, large-scale engineering projects.

References 1

- [1] Amazon.com: Swaytail Premium Large Metal 4WD shock absorption robot tank car chassis kit for Arduino/Raspberry Pie/MicroBit with 33 DC motor, shock absorbing tracked crawler with damping system DIY Maker Model : Toys & Games, <https://www.amazon.com/Premium-Absorption-Raspberry-MicroBit-Absorbing/dp/B09CYDB8H4> (accessed Nov. 19, 2025).
- [2] Amazon.com: For Raspberry Pi 5 8GB starter kit includes with Raspberry Pi 5 8GB, PI5 case with active cooler, 27W USB type-C power supply, 64GB SD card and TF Card Reader, HDMI cable, ethernet cable (8 items) : Electronics, <https://www.amazon.com/Raspberry-Starter-Active-Cooler-Ethernet/dp/B0CXXFFYNB> (accessed Nov. 19, 2025).
- [3] Walmart.com, L293D Motor Drive Shield Expansion Board For Arduino Uno Mega2560 & Duemilanove, Envistia Mall, <https://www.walmart.com/ip/L293D-Motor-Drive-Shield-Expansion-Board-For-Arduino-Uno-MEGA2560-Duemilanove/400667780?wmlspartner=wlp&selectedSellerId=101028229> (accessed Apr. 12, 2025).
- [4] "4D Lidar L1 application Scenarios_4D LIDAR L1 efficacy: Unitree Robotics," 4D Lidar L1 Application Scenarios_4D Lidar L1 Efficacy | Unitree Robotics, <https://www.unitree.com/LiDAR/> (accessed Nov. 19, 2025).
- [5] Amazon.com: Arducam IMX219 8MP Autofocus USB Camera Module, 3280(H)x2464(V)@30fps with Single Microphone for Linux, Windows, and Mac OS, <https://a.co/d/0fThY8IZ> (accessed Apr. 4, 2026).
- [6] Amazon.com: CJRSLRB® Gas Detection Module MQ-2 MQ-3 MQ-4 MQ-5 MQ-6 MQ-7 MQ-8 MQ-9 MQ-135 sensor module each of them 1pcs total 9pcs sensor kit : Tools & Home Improvement, <https://www.amazon.com/CJRSLRB-Detection-Module-MQ-135-Sensor/dp/B016KABTDK> (accessed Nov. 19, 2025).
- [7] Mavdrones, "Tattu 2200 mah lipo battery with XT90-S Anti-Spark connector," Mavdrones, <https://www.mavdrones.com/product/tattu-22-2v-25c-6s-2200-mah-lipo-battery-with-xt90-s-anti-spark-connector/> (accessed Nov. 19, 2025).

References 2

- [8] Amazon.com: BrosTrend 650mbps linux compatible WIFI adapter supports Kali Linux, ubuntu, Mint, debian, kubuntu, zorin, pureos, Raspberry Pi 2+, windows, dual band USB Wireless Adapter W/ long range WIFI antenna : Electronics, <https://www.amazon.com/BrosTrend-600Mbps-Wireless-Internet-AC2/dp/B01GC8XH0S> (accessed Nov. 19, 2025).
- [9] E. Ackerman, "Boston Dynamics' Spot Robot Dog now available for \$74,500," IEEE Spectrum, <https://spectrum.ieee.org/boston-dynamics-spot-robot-dog-now-available> (accessed Nov. 19, 2025).
- [10] Base Year:2024], "Autonomous Mobile Robots Market Size (\$13.7 billion) 2030," STRATEGIC MARKET RESEARCH, <https://www.strategicmarketresearch.com/market-report/autonomous-mobile-robots-market> (accessed Nov. 19, 2025).
- [11] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, and F. Zhang, "Point-LIO: Robust High-Bandwidth Light Detection and Ranging Inertial Odometry," *Advanced intelligent systems*, vol. 5, no. 7, Apr. 2023, doi: <https://doi.org/10.1002/aisy.202200459>.
- [12] Adam Zewe | MIT News, "Engineers enable a drone to determine its position in the dark and indoors," MIT News | Massachusetts Institute of Technology, <https://news.mit.edu/2025/engineers-enable-drone-determine-its-position-dark-and-indoors-0213> (accessed Nov. 19, 2025).
- [13] "What is a Lithium Polymer Battery? | LiPo," Ossila, 2021. <https://www.ossila.com/pages/lithium-polymer-battery>
- [14] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot Operating System 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, 2022, doi: <https://doi.org/10.1126/scirobotics.abm6074>.
- [15] "ROS 2 Documentation: Jazzy documentation," ROS 2 Documentation, <https://docs.ros.org/en/jazzy/index.html> (accessed Sept. 2025).
- [16] "Nav2," Navigation2 Documentation, <https://docs.nav2.org/> (accessed Jan. 2026).
- [17] "Costmap 2D," Navigation2 Documentation, <https://docs.nav2.org/configuration/packages/configuring-costmaps.html> (accessed Jan. 2026).
- [18] "RViz," ROS 2 Documentation: Jazzy documentation, <https://docs.ros.org/en/jazzy/Tutorials/Intermediate/RViz/RViz-Main.html> (accessed Sept. 2025).
- [19] Point Cloud Library (PCL), PointClouds.org, <https://pointclouds.org/> (accessed Apr. 1, 2026).
- [20] "ROS 2 Jazzy Jalisco Released!," Open Robotics Discourse, <https://discourse.openrobotics.org/t/ros-2-jazzy-jalisco-released/37862> (accessed Sept. 2025).



**THANK YOU
&
QUESTIONS**